AI-powered Cloud Masking Report

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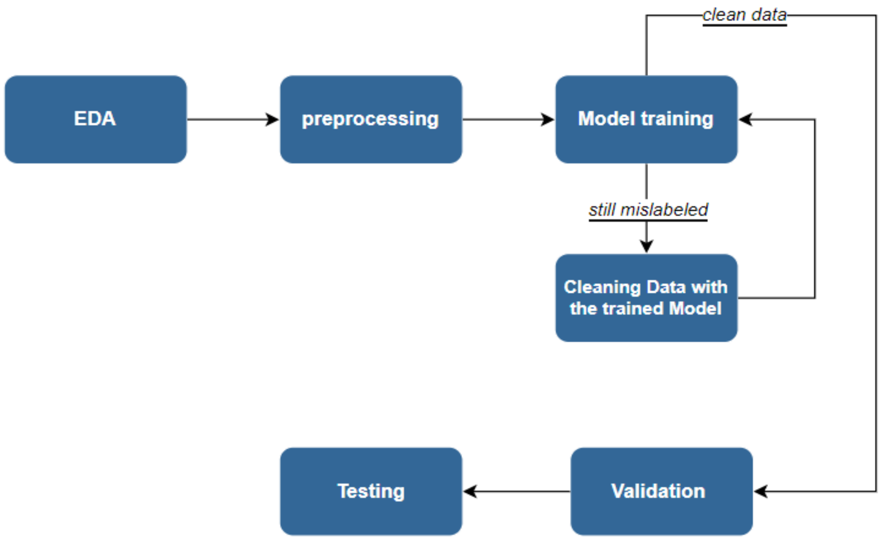
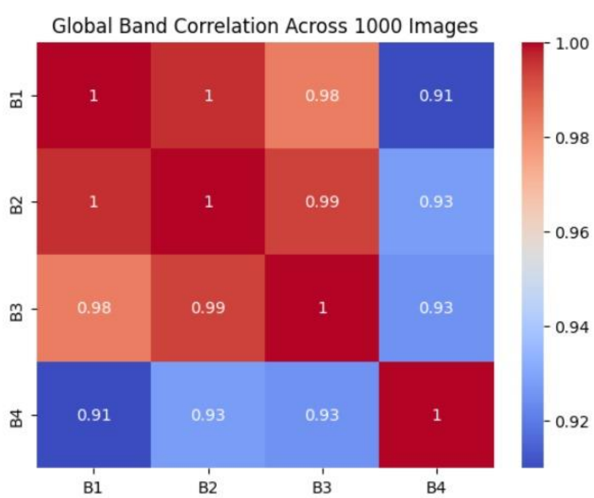
Project Pipeline:

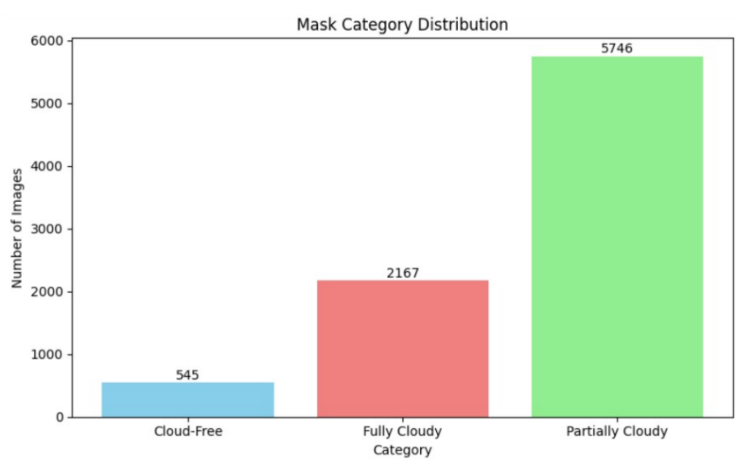


📊 Exploratory Data Analysis:

In our exploratory data analysis, we visualized random images from dataset with masks to uncover key insights, which are summarized below:

1. **Cloud Coverage Distribution**: The analysis revealed the distribution of cloud coverage, categorized as fully clouded, partially clouded, and cloud-free, providing a clear understanding of the dataset's composition.
2. **Data Labeling Issues**: A significant number of mislabeled data points were identified, indicating potential challenges in data quality.
3. **Prevalence of Mislabeled Data**: The majority of mislabeling occurred in the fully clouded.
4. **Correlation Analysis**: By applying a heatmap, we observed a high correlation (close to 1) across all bands.





🛠️ Preprocessing Module:

To enhance the dataset for model training, we implemented the following preprocessing steps:

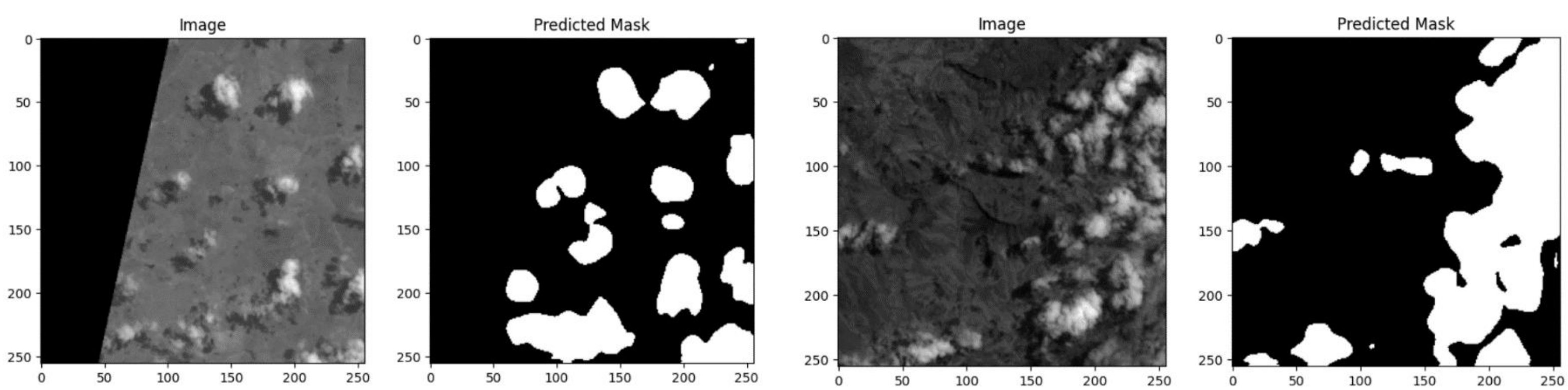
1. **Dimensionality Reduction**: Principal Component Analysis (PCA) was applied to bands 3 and 4, effectively reducing them to a single channel while preserving critical information.
2. **Data Augmentation**: To expand the training dataset and improve model robustness, we augmented the images using:
   - Gaussian noise, Horizontal and vertical flipping, and Rotation.
   - We explored additional techniques such as shearing, shifting, and zooming but excluded them as they distorted cloud shapes, compromising data integrity.

🤖 Model Selection & Training Module

1. Deep Learning Model:

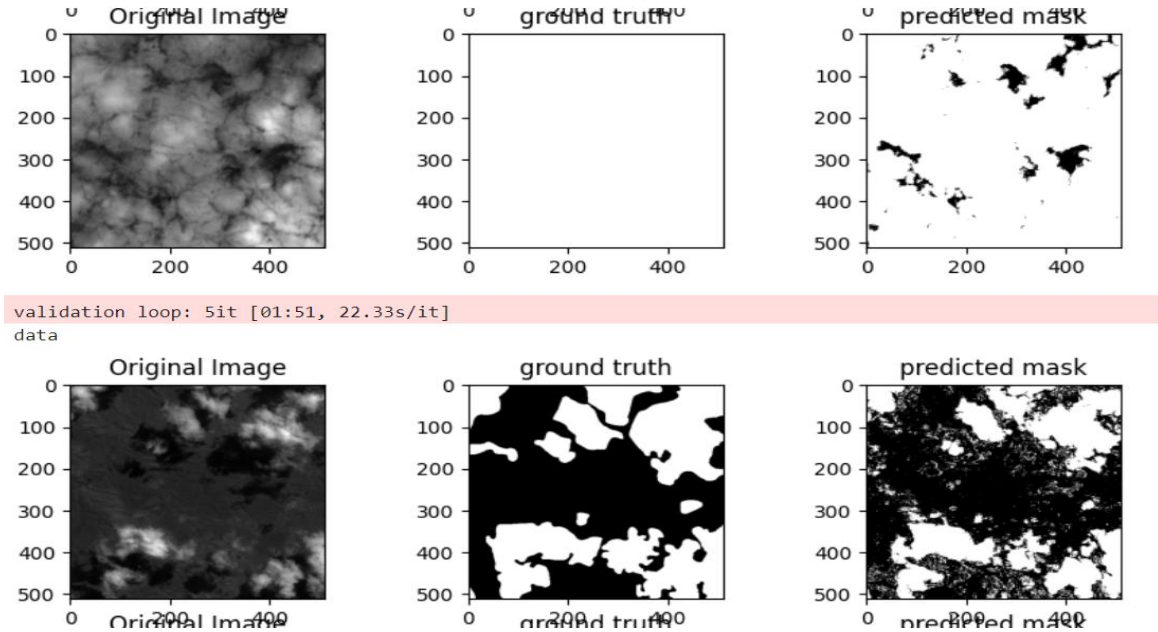
To develop an effective model for our task with dice score **0.86**, we implemented the following steps in our deep learning pipeline:

1. **Initial Model Training**:
   - We trained a UNet-Efficient model on the training dataset, excluding fully clouded and cloud-free samples due to the high prevalence of mislabeling in the fully clouded category.
2. **Data Cleaning**:
   - The trained UNet-Efficient model was used to clean the entire dataset, removing mislabeled data and improving overall data quality.
3. **Training on Cleaned Data**:
   - A new model was trained on the cleaned dataset to leverage the improved data quality for better performance.
4. **Architecture Exploration**:
   - We evaluated multiple architectures, including UNet, UNet++, Residual UNet, and DeepLab. Comparative analysis showed no significant differences in Dice coefficient scores across these models. Consequently, we selected UNet as the final architecture due to its simplicity.



2. **Machine Learning Model:**

   We evaluated XGBoost for cloud segmentation, achieving a Dice score of **0.66**. While it captured some cloud details, it produced noisy masks due to its reliance on edge-based features. Deep learning models were preferred for their superior performance and robustness.



Performance Analysis Module:

1. **Evaluation Metric**:
   - The Dice coefficient was employed as the primary metric for the model's accuracy.
2. **Loss Function**:
   - A composite loss function was used, defined as:
   - 0.5*(binary cross entropy error) +0.5*(Dice loss)
3. **Error Analysis**:
   - Visualization of model errors revealed the following limitations:
     - **Fine Detail Capture**: The model struggled to accurately segment fine details, such as the thin boundaries of clouds.
     - **Small Non-Cloud Regions**: In cases where masks were almost white (indicating clouds) with small non-cloud areas, the model failed to detect these minority regions effectively.

🚀 Enhancement & Future Work:

1. **Explainable AI**:
   - Use Grad-CAM or SHAP to visualize and explain model predictions.
2. **External Datasets**:
   - Fine-tune using larger cloud segmentation datasets for better generalization.
3. **Temporal Modeling**:
   - Integrate Transformers to track cloud dynamics over time.

📋 Workload:

Name	Nesma Abd-El-Kader	Eman Ibrahim	Sara Gamal	Yousef Osama
Workload	**DL** **Data cleaning**	**DL** **Data cleaning**	**EDA** **Preprocessing**	**ML**